

Developing a robust and efficient methodology for fMRI time-series clustering

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Workplan Contribution

This project contributed to data analysis (WP4-5) and systems technology (WP6-7). In particular, the project aligns with Workplan 7: “Advances in neuromorphic computing & imaging technologies”. Overall, the current project developed a computationally efficient clustering approach for large datasets and translated the algorithm back to neuroscience research to address the question of cortical and subcortical organisation over time.

Overview

Machine-learning technologies are gradually becoming an indispensable tool in data analysis. Moving beyond linear relationships, these tools enable us to model complex non-linear interactions and generalise uncovered relationships to new observations, thereby creating more accurate world models. As such, there is a worldwide effort to create accessible, easy-to-use machine-learning tools and translate them to various socio-economic sectors. However, there are significant barriers to entry to effectively utilising machine learning. A fundamental problem is related to the computational power required to train a model. Given that modelling these complex relationships requires employing computer clusters across extensive training periods, the cost of utilizing machine learning can be prohibitive. This sentiment is echoed by low-tech proponents of scientific research, who discuss the feasibility of applying and reproducing machine-learning models in academic and industrial settings that lack the necessary infrastructure. The cost is further amplified by the fact that, in machine-learning practice, researchers tend to develop multiple competing models and examine their performance in unseen data. The benefits of this approach are clear: we can identify the best-performing model by searching across a potential cohort. However, this approach multiplies the financial and time costs of machine learning. In addition to that, it does not allow us to examine why different models perform better and in what manner. The present project, developed within the context of the “NeuronsXnets” Program, aimed to create a computationally efficient machine-learning algorithm and subsequently use it to examine the fundamental properties of cortical and subcortical connectivity organisation.

State of the art in clustering analysis

Unsupervised learning is a branch of machine learning that focuses on finding hidden patterns and structures in data without relying on predefined labels. It enables researchers to uncover relationships, groupings, or representations that may not be immediately apparent, making it especially powerful in exploratory data analysis (Ikotun et al., 2023). Within this paradigm, clustering emerges as a key method, where data points are grouped based on similarity, enabling the identification of distinct substructures in large and complex datasets (Figure 1).

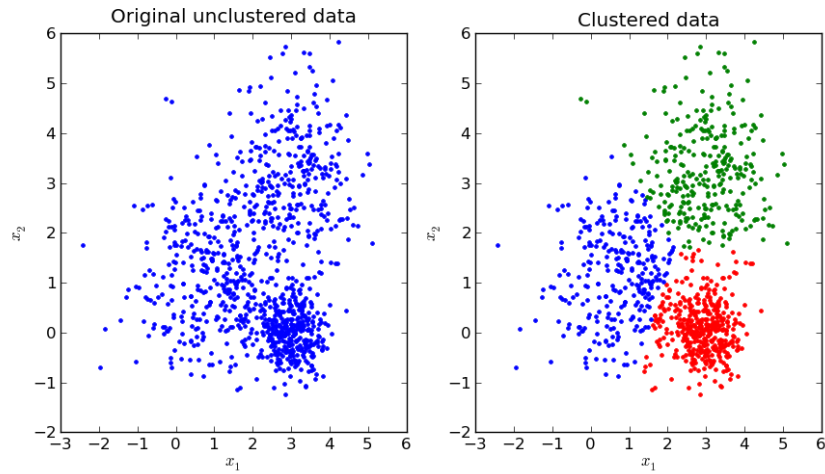


Figure 1. Examination of a clustering approach to data analysis. Unlabelled data (left column) are fed to the clustering algorithm. Then, each datapoint is assigned to a discrete cluster.

In collaboration with the Neurolingo software company, we attempted to create variants of well-known clustering algorithms (KMeans, KMedoids) that minimise computational load, thereby achieving faster execution while requiring fewer memory resources. This improved infrastructure enables us to create a comprehensive comparison of algorithm performance in fMRI timeseries.

Application in Neuroscience

Clustering methodologies are especially relevant in neurosciences, where we need to establish commonalities in large cohorts of data, such as EEG (changes in the electrical signal produced by populations of neurons) and fMRI signals (changes in the blood oxygenation level -also known as BOLD signal). While time series data are inherently noisy, clustering approaches can uncover fundamental organisational properties. We can then leverage these organisational properties and link brain timeseries data back to behaviour.

So far, clustering has been primarily applied in the context of fMRI functional connectivity (FC). During rest, fMRI BOLD timeseries across different brain regions display spontaneous co-fluctuations, termed functional connectivity (Biswal et al., 1995). Apart from static approaches, such as estimating FC over a specified time period, we can estimate FC at every sampling point, yielding an estimate of time-varying FC. Then, we can apply clustering to reduce the complex time-resolved FC maps to representative connectivity patterns. These patterns capture distinct features of functional connectivity organisation, such as arousal, structural connections, and long- and short-range cortical communications (Demertzi et al., 2019; Mortaheb et al., 2022). These patterns have been influential across different research lines. One line has attempted to differentiate wakefulness from unconscious state (anaesthesia, disorders of consciousness) based on how patterns evolve (Castro et al., 2024; Demertzi et al., 2019). Another line attempts to link pattern presence to specific subjective and attentional mental content (Mortaheb et al., 2022; Turker et al., 2024). Therefore, these patterns hold explanatory value for subject experience.

Current approaches apply K-Means clustering to a set of time-varying FC data and then link the patterns to behaviour. This approach faces specific caveats. Firstly, depending on the data pre-processing schemes, the dataset size can range from a few hundred megabytes to hundreds of gigabytes. This can inflate both computational cost and computation time. This issue is further exaggerated by the recent practice of permuting across possible preprocessing parameters to ensure the robustness of the methodology (Luppi et al., 2024). For example, a typical analysis will examine the effect of fMRI global signal regression (with two options), brain parcellation scheme (with five options), and number of clusters (with six options), resulting in a total of 60 different models. Finally, there is no consensus that different clustering distances, such as Euclidean or Manhattan distance, produce similar results.

The present project aims to address these issues simultaneously. By developing a fast and efficient clustering software, we can examine the degree to which clustering is robust and informative for brain connectivity organisation.

Internship work

During my time at Neurolingo, I was responsible for data organisation and preparation, clustering, benchmarking, and algorithm interpretation. My goal was to assess the effectiveness of the developed clustering software on neuroimaging data.

Dataset description.

We utilised a large resting state dataset consisting of a 20-minute resting state fMRI recording across 285 participants (~60GB). The dataset was initially described in Lumaca et al. (2024).

Algorithm benchmarking: time efficiency

We utilised the validated MiniBatchesKMeans algorithm (Sculley, 2010) from the Python scikit-learn package (Pedregosa et al., 2011) as the benchmark of time performance. The mini-batch implementation of K-Means is a standard practice when loading the whole dataset into computer memory is not feasible. Overall, our implementation significantly outperformed MiniBatchesKMeans, both in overall performance and in every specific cluster examined (mean MiniBatchesKMeans time = 205 seconds, mean developed software = 10 seconds) (Figures 2,3).

Algorithm benchmarking: spatial similarities

Having validated that the developed software significantly outperforms the existing one, we examined whether the software yielded comparable clusters. Across clusters and preprocessing parameters, we demonstrate that the developed software yields similar clustering topologies, indicating no loss in clustering performance (Figure 4).

Connectivity topologies differ across clustering distances.

Given the time and spatial efficiency of our algorithm, we set out to examine whether the clustering approach is robust to the distance metric. The distance metric reflects our assumptions about how fMRI time series can be similar. Therefore, different distances might yield different results. Our results suggest stark differences across distance metrics, potentially because different algorithms capture different properties of functional connectivity organisation (Figure 5).

Optimal algorithm clustering

Finally, we set out to examine whether there is convergence in the optimal amount of brain connectivity patterns uncovered. While we were able to find that 6-7 clusters contained a maximal amount of differentiable yet informative clusters, cosine distance did not reach convergence.

Conclusions and future directions

In the present internship, we aimed to develop an efficient computation tool to perform clustering and then apply this tool to compare different approaches to fMRI timeseries clustering. Our results suggest that both preprocessing parameters and clustering distance have a significant impact on the identified brain connectivity patterns. While we provide a systematic evaluation of specific clustering parameters, our results raise concerns about the comparability of the clustering approach across different studies.

Moving forward, we aim to provide a complete account of how clustering can help us understand time-varying FC. Based on the tools and analysis strategies developed within the context of the “NeuronsXnets” collaboration, we aim to extend our results to various datasets across different consciousness states and experimental conditions, providing an optimal analysis pipeline for meaningful and interpretable brain dynamics.

Annex

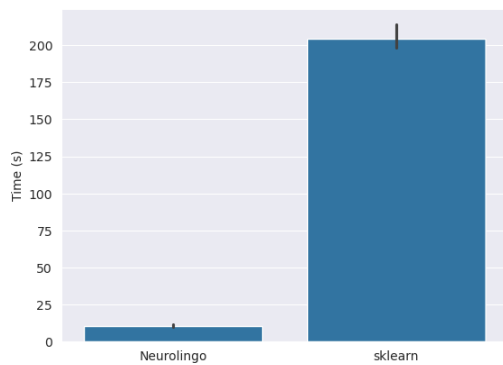


Figure 2. Cross-cluster time performance

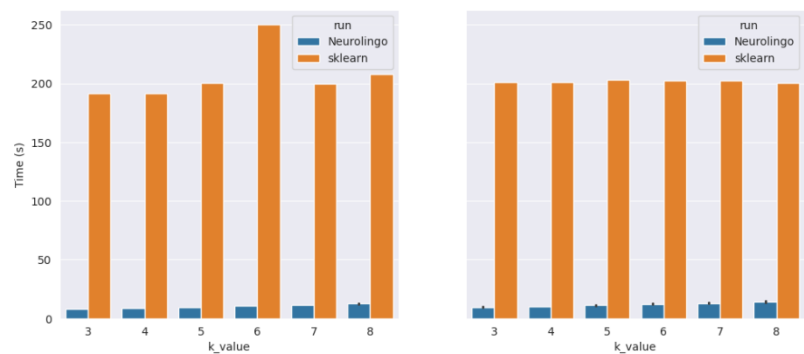


Figure 3. Time performance stratified by cluster

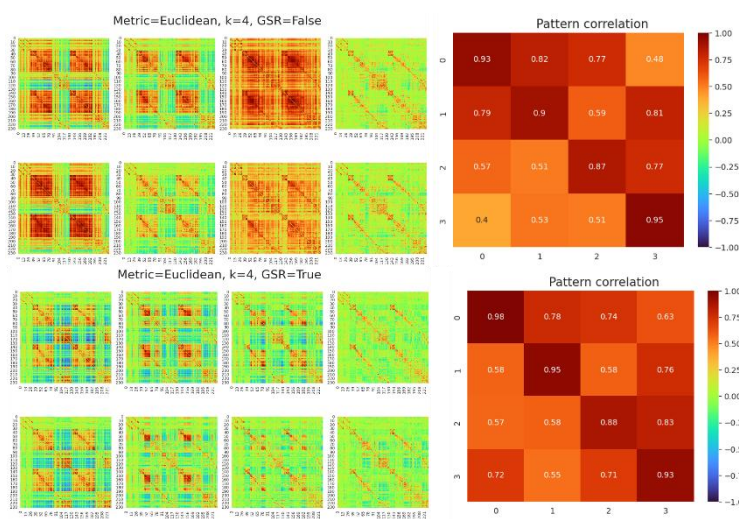


Figure 4. Comparison of clustering performance.

Left image: When the global signal was not regressed, we observed similar topologies in the observed cluster. Right image: Similarly, when the global signal was regressed, similar clusters were present. Top rows reflect the clusters obtained from the MiniBatchKMeans algorithm, bottom rows reflect the clusters obtained from the developed software.

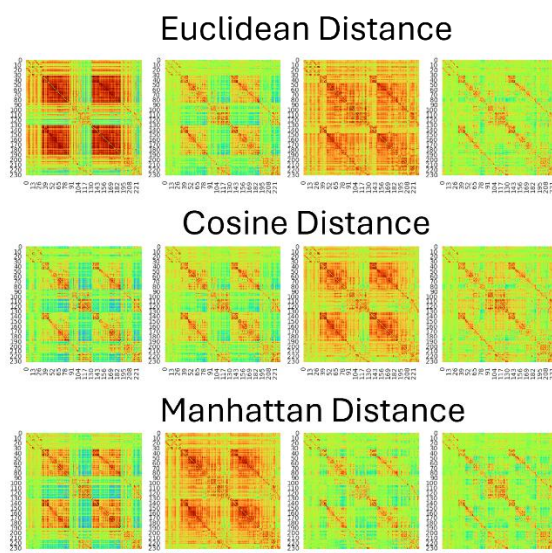


Figure 5. Clustering distance significantly affects identified brain connectivity patterns

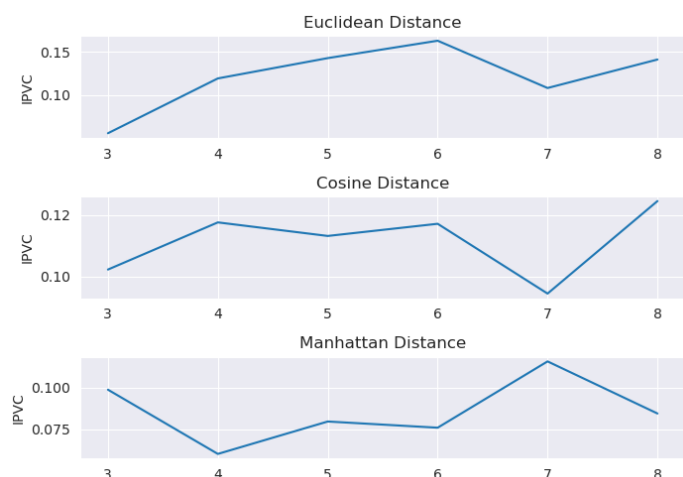


Figure 6. Clustering distance significantly affects the optimal number of informative clusters

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